

Maximum Entropy Transport Plan

Let $r, c \in \Sigma_d$, and define

$$U(r, c) = \left\{ P \in \mathbb{R}_+^{d \times d} : P\mathbf{1} = r, \quad P^\top \mathbf{1} = c \right\}.$$

We want to show that, for every $P \in U(r, c)$,

$$h(P) = h(r) + h(c) \iff P = rc^\top.$$

First, we prove that $rc^\top \in U(r, c)$. Let $i \in \{1, \dots, d\}$. Then

$$\sum_j r_i c_j = r_i \sum_j c_j = r_i \cdot 1 = r_i.$$

Similarly, for every $j \in \{1, \dots, d\}$,

$$\sum_i r_i c_j = c_j \sum_i r_i = c_j \cdot 1 = c_j.$$

Therefore,

$$rc^\top \in U(r, c).$$

Now suppose that

$$P = rc^\top.$$

Then

$$\text{KL}(P \| rc^\top) = \text{KL}(P \| P) = \sum_{i,j} P_{ij} \log \left(\frac{P_{ij}}{P_{ij}} \right) = \sum_{i,j} P_{ij} \log(1) = 0.$$

Using the identity

$$\text{KL}(P \| rc^\top) = h(r) + h(c) - h(P),$$

see the derivation here : proof 1
we obtain

$$0 = h(r) + h(c) - h(P).$$

Hence,

$$h(P) = h(r) + h(c).$$

Conversely, suppose that

$$h(P) = h(r) + h(c).$$

Then, using again

$$\text{KL}(P \| rc^\top) = h(r) + h(c) - h(P),$$

we get

$$\text{KL}(P \| rc^\top) = 0.$$

Recall that the Kullback–Leibler divergence is nonnegative, and that

$$\text{KL}(P \| rc^\top) = 0 \iff P = rc^\top.$$

Therefore,

$$P = rc^{\top}.$$

Thus, for every $P \in \mathcal{U}(r, c)$,

$$h(P) = h(r) + h(c) \iff P = rc^{\top}.$$

In particular, the unique element of $\mathcal{U}(r, c)$ maximizing the entropy is

$$P = rc^{\top}.$$